

Electrochemistry

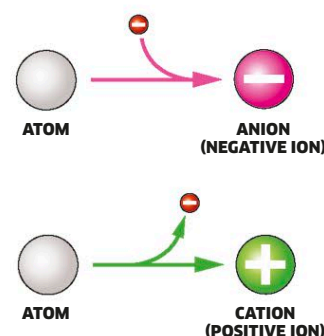
Electricity and chemical reactions are closely linked, and together fall under the heading electrochemistry. Electrochemistry is the study of chemical processes that cause electrons to move.

An electric current is a steady flow of electrons, the tiny negative particles that whizz around in the shells of atoms. Electrons can flow in response either to a chemical reaction taking place inside a battery, or to a current delivered by the main electrical grid.

Electricity is key to electrolysis. This process is used in industries to extract pure elements from ionic compounds (see p.44) that have been dissolved in a liquid known as an electrolyte. Electrolysis can also be used to purify metals, and a similar process can be used to plate (cover) objects with a metal. The result depends on the choice of material of the electrodes and, in particular, the exact contents of the electrolyte.

Ions and redox reactions

Chemical reactions where electrons are transferred between atoms are called oxidation-reduction (redox) reactions. Atoms that have lost or gained electrons become ions, and are electrically charged. Atoms that gain electrons become negative ions (anions). Atoms that lose electrons become positive ions (cations). These play an important role in electrolysis.

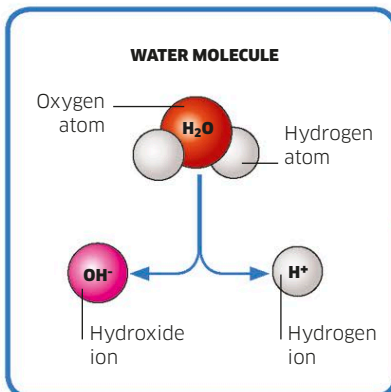


Electrolysis

Ionic compounds contain positive and negative ions. They can be separated using electricity, by a process called electrolysis. If electricity passes through an electrolyte (an ionic compound that has been dissolved in water), the negative ions in the electrolyte will flow toward the positive electrode and the positive ions will flow toward the negative electrode. The products created in the process will depend on what is in the electrolyte. This diagram shows how water (H_2O) can be split back into its original pure elements, oxygen and hydrogen. The two gases can be trapped and collected as they bubble up along the electrodes.

Water as electrolyte

The electric current makes each neutral water molecule (H_2O) split up into electrically charged ions: a positive hydrogen ion (H^+) and a negative hydroxide ion (OH^-).



3 The oxygen atoms bond (see p.17) to form oxygen gas molecules (O_2).

2 The negative hydroxide ions (OH^-) are attracted to the positive electrode, where they lose electrons and become neutral oxygen atoms.

1 In the electrolyte, water molecules split up into ions as current passes through.

Battery
The battery has a negative (-) and a positive (+) terminal.

Positive electrode
This positive electrode, known as an anode, is made of platinum, a metal.

Electric current
As electrons start flowing through the electrodes and back toward the positive terminal, an electric current runs through the whole apparatus.

Negative electrode
This negative electrode, known as a cathode, is also made of platinum.

Electrolyte
This electrolyte is just water (H_2O). It conducts electricity generated by the flow of electrons.

3 The hydrogen atoms bond to form hydrogen gas molecules (H_2).

2 The positive hydrogen ions (H^+) are attracted to the negative electrode, where they gain electrons and become neutral hydrogen atoms.

1 In the electrolyte, water molecules split up into ions as current passes through.

